

AI-Powered Digital Twin of India's Climate

Physics & Mathematics-Informed Architecture and Implementation Plan

ISRO Hackathon Submission | Domain: Climate Resilience & Disaster Preparedness

Table of Contents

- [1. Problem Formulation](#)
- [2. Physical Governing Equations](#)
- [3. System Architecture \(4 Layers\)](#)
- [4. Mathematical Formulations per Layer](#)
- [5. AI Model Architecture Details](#)
- [6. Advanced Research Modules](#)
- [7. Data Schema & Infrastructure](#)
- [8. Technology Stack](#)
- [9. Phased Implementation Plan](#)
- [10. Verification & Benchmarking](#)

1. Problem Formulation

1.1 Scientific Objective

Build a **near-real-time, interactive Digital Twin** of India's climate that:

- Fuses heterogeneous observational data (satellite + ground) into a physically consistent model state.
- Uses AI surrogate models to replace computationally expensive NWP steps.
- Provides physically-constrained "what-if" scenario simulation in milliseconds.

1.2 Domain Scope

Parameter	Value
Spatial domain	Indian subcontinent: 6°N–37°N, 68°E–97°E
Target resolution	4 km × 4 km (INSAT native); upscaled to 1 km for urban tiles
Temporal resolution	Half-hourly assimilation cycle; daily forecast

Parameter	Value
	rollout
Historical depth	IMD records 1901–present (~125 years)
Primary variables	Rainfall (RF), T_max, T_min, LST, Soil Moisture (SM), 500 hPa Wind Field

2. Physical Governing Equations

This section defines the exact physics the system must respect. Every AI component is constrained by these laws.

2.1 Atmospheric Dynamics — Primitive Equations (Shallow Atmosphere)

The dry, hydrostatic primitive equations on pressure coordinates (p):

Momentum (u, v):

$$\begin{aligned} du/dt &= -(u \cdot du/dx + v \cdot du/dy + w \cdot du/dp) + fv - (1/\rho) \cdot dp/dx + F_u \\ dv/dt &= -(u \cdot dv/dx + v \cdot dv/dy + w \cdot dv/dp) - fu - (1/\rho) \cdot dp/dy + F_v \end{aligned}$$

Continuity (Mass Conservation):

$$du/dx + dv/dy + dw/dp = 0$$

Thermodynamic Energy:

$$dT/dt = -(u \cdot dT/dx + v \cdot dT/dy) + (w/c_p) \cdot (R \cdot T/p - dT/dp) + Q/c_p$$

where f = Coriolis parameter, w = vertical velocity in pressure coords, Q = diabatic heating, c_p = specific heat of dry air at constant pressure.

2.2 Moisture Transport — Advection-Diffusion Equation

For a scalar tracer q (specific humidity or pollutant concentration):

$$dq/dt + u \cdot dq/dx + v \cdot dq/dy + w \cdot dq/dp = K_h \cdot \text{Laplacian}(q) + S_q$$

where K_h = horizontal diffusivity coefficient, S_q = source/sink term (evaporation, condensation).

This PDE is embedded directly into the PINN loss function.

2.3 Land Surface Energy Balance

The net energy flux at the surface:

$$R_n = H + LE + G$$

- R_n = Net radiation = $(1-\alpha) \cdot R_{s_down} + R_{L_down} - \epsilon \cdot \sigma \cdot T_s^4$

- $H = \text{Sensible Heat Flux} = \rho \cdot c_p \cdot (T_s - T_a) / r_{ah}$
- $LE = \text{Latent Heat Flux} = \rho \cdot c_p \cdot (e_s - e_a) / (\gamma \cdot r_{aw})$
- $G = \text{Ground Heat Flux (residual)}$

LST from INSAT-3DR provides T_s , which feeds sEKF to correct H and soil moisture θ .

2.4 Hydrostatic Balance (Vertical Structure)

$$d\Phi/d(\ln p) = -R_d \cdot T_v$$

where Φ = geopotential, R_d = gas constant for dry air, T_v = virtual temperature.

2.5 Saturation & Condensation (Clausius-Clapeyron)

$$de_s/dT = L_v \cdot e_s / (R_v \cdot T^2)$$

Used to derive convective instability thresholds for cloudburst/flash-flood triggers.

2.6 Soil Moisture Breakpoint — Hypersensitive Regime [NEW]

Soil moisture does **not** influence evapotranspiration linearly. Below the permanent wilting point θ_{pwp} , the soil-to-atmosphere water pathway collapses entirely:

$$LE(\theta) = LE_{pot} \cdot \min(1, \max(0, (\theta - \theta_{pwp}) / (\theta_{fc} - \theta_{pwp})))$$

- θ_{fc} = field capacity (0.30 m³/m³ for loam), ~~θ_{pwp} = permanent wilting point~~ (0.14 m³/m³)
- When $\theta < \theta_{pwp}$: $LE \rightarrow 0$, ALL of R_n goes to $H \Rightarrow$ Sensible Heat Flux spike

Hypersensitive Regime Indicator (HRI):

$$HRI(x,y,t) = \begin{cases} 1 & \text{if } \theta(x,y,z,t) < \theta_{pwp} \text{ for } z \text{ in } \{\text{layer1, layer2, layer3}\} \\ 0 & \text{otherwise} \end{cases}$$

The critical insight from *Journal of Climate* research is that the **3D vertical structure** of SM errors — not just the surface layer — drives uncertainty in subseasonal heatwave onset. The sEKF state vector is therefore extended to 3 soil layers:

$$x_{\text{extended}} = [T_s, \theta_1, \theta_2, \theta_3, H, LE, HRI]^T$$

A $\theta < \theta_{pwp}$ event in **all three layers simultaneously** triggers the hypersensitive heatwave flag in the dashboard.

3D Error Covariance Extension: The background error covariance B_{soil} in the ETKF is expanded from 2D surface fields to a full 3D vertical soil profile using a separable formulation:

$$\begin{aligned} B_{\text{soil}}(z_1, z_2) &= B_{\text{horiz}}(x,y) \cdot C_{\text{vert}}(z_1, z_2) \\ C_{\text{vert}}(z_1, z_2) &= \exp(-|z_1 - z_2| / L_z) \quad [L_z \sim 0.5\text{m, exponential vertical correlation}] \end{aligned}$$

This captures how a deep drought propagates upward to trigger surface sensible heat anomalies.

2.7 Aerosol-Lightning-Rainfall Non-Linear Coupling [NEW]

Aerosol Optical Depth (AOD) modifies cloud droplet effective radius r_e via Twomey's relation:

$$r_e \propto N_d^{-1/3} \quad \text{where} \quad N_d = f(\text{AOD}, \text{RH}, \text{CCN_activation})$$

- High AOD => high cloud droplet number N_d => smaller r_e => suppressed warm rain, enhanced ice formation
- The resulting enhanced updraft-ice interaction drives positive cloud-to-ground (CG+) lightning:

Lightning Flash Rate Parameterization (Price-Rind adapted):

$$F_{\text{lightning}} = a \cdot W_{\text{max}}^b \cdot g(\text{AOD}, \text{RH})$$
$$g(\text{AOD}, \text{RH}) = \exp(c \cdot \text{AOD}) \cdot \text{sigmoid}((\text{RH} - \text{RH}_0) / \text{sigma_RH})$$

where W_{max} = maximum updraft velocity, a , b , c = empirical constants calibrated from IITM lightning data, $\text{RH}_0 \sim 70\%$, $\text{sigma_RH} = 10\%$.

The non-linear AOD-RH coupling: At $\text{RH} < 60\%$, aerosols primarily scatter radiation with minimal cloud activation. At $\text{RH} > 70\%$, the same AOD produces dramatic CCN activation, invigoration, and CG+ lightning bursts. This bifurcation is embedded as a conditional gate in the PINN.

2.8 ATAL Radiation Budget Forcing [NEW]

The Asian Tropopause Aerosol Layer (ATAL) at 100–200 hPa in the UTLS traps anthropogenic pollutants within the monsoon anticyclone. Its forcing on the top-of-atmosphere (TOA) radiation budget:

ATAL Shortwave Forcing:

$$dF_{\text{SW}} = -\text{AOD_ATAL} \cdot (1 - f_c)^2 \cdot S_0/4 \cdot (1 - R_s)^2 \cdot \text{beta}_s$$

ATAL Longwave Forcing:

$$dF_{\text{LW}} = +\text{epsilon_ATAL} \cdot \text{sigma} \cdot (T_{\text{ATAL}}^4 - T_{\text{eff_cloud}}^4)$$

where epsilon_ATAL = ATAL emissivity (function of aerosol column loading), $T_{\text{ATAL}} \sim 195\text{K}$ (UTLS temperature).

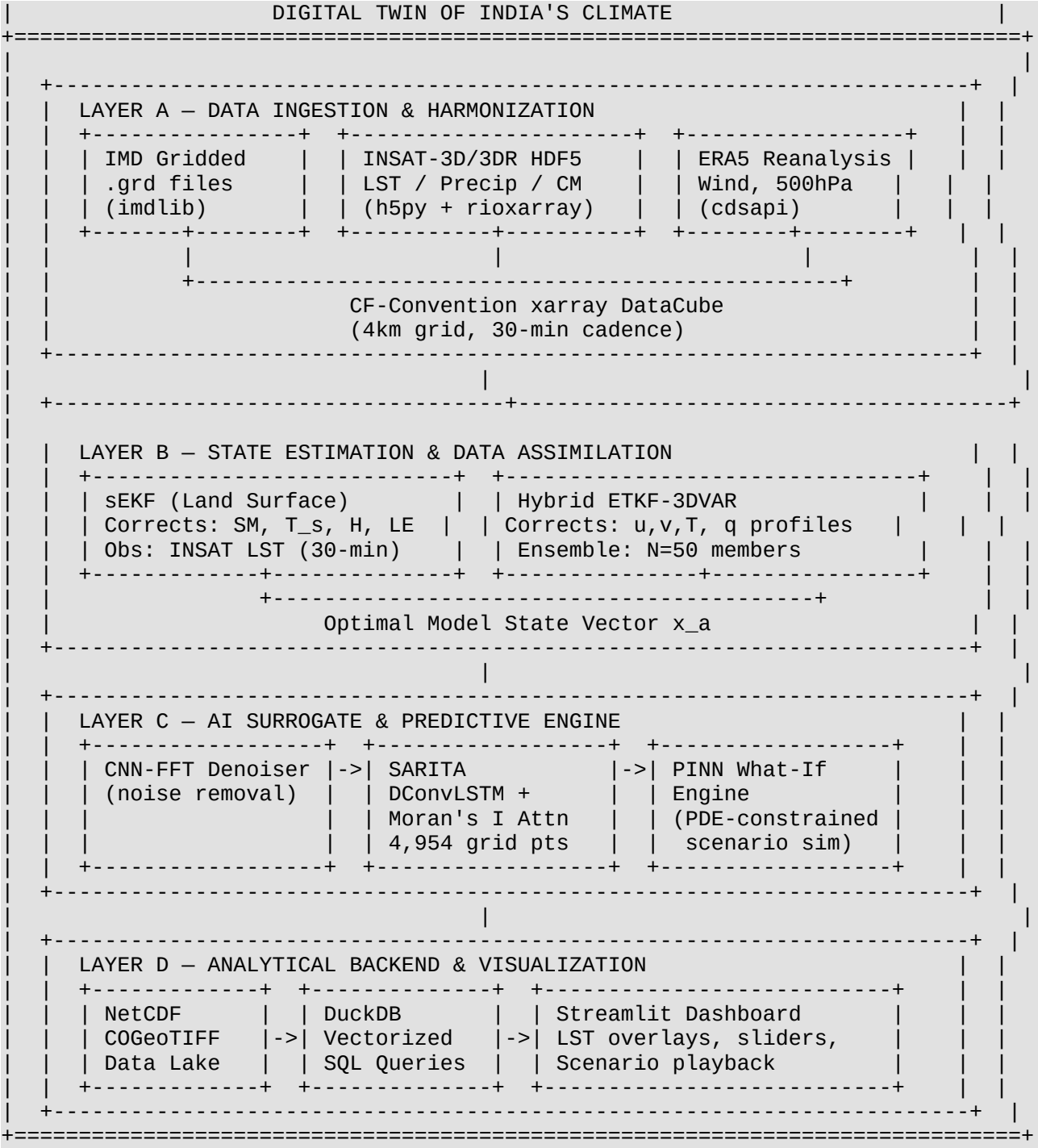
Net ATAL forcing modifies the surface energy balance as a boundary condition:

$$R_{\text{n_surface}} = R_{\text{n_clear}} - dF_{\text{SW}} - dF_{\text{LW}}$$

This is assimilated using INSAT-3D sounder brightness temperatures at $6.2\mu\text{m}$ (water vapour channel) to derive UTLS temperature profiles and infer epsilon_ATAL .

3. System Architecture

+=====+



4. Mathematical Formulations per Layer

Key upgrade: Sections 4.2, 4.3, and 4.4 now integrate all 5 advanced research modules. Each is marked [NEW] at point of insertion.

4.1 Layer A — Data Harmonization

Reprojection of INSAT geostationary (GEOS) to Mercator (geographic lat/lon). Implemented via `rioxarray.reproject()` using `EPSG:4326` as target CRS.

Regridding INSAT 4km to IMD 27km uses **bilinear interpolation**:

$$f(x,y) = (1/(Dx \cdot Dy)) \cdot [f(x1,y1) \cdot (x2-x)(y2-y) + f(x2,y1) \cdot (x-x1)(y2-y) + f(x1,y2) \cdot (x2-x)(y-y1) + f(x2,y2) \cdot (x-x1)(y-y1)]$$

Missing Value Imputation for IMD .grd fill (-999.0) uses **Kriging** (Gaussian variogram):

$$\text{gamma}(h) = C_0 + C_1 \cdot [1 - \exp(-h^2/a^2)]$$

where h = lag distance, C_0 = nugget, C_1 = sill, a = range.

4.2 Layer B — Data Assimilation

4.2.1 Simplified Extended Kalman Filter (sEKF) — Land Surface [UPGRADED]

Extended state vector (now 3 soil layers + HRI flag per §2.6):

$$x = [T_s, \text{theta_1}, \text{theta_2}, \text{theta_3}, H, LE, HRI]^T$$

Prediction step:

$$\begin{aligned} x_hat_{\{k|k-1\}} &= f(x_{\{k-1\}}) && \text{[NOAH-LSM 3-layer nonlinear forward model]} \\ P_{\{k|k-1\}} &= F_k \cdot P_{\{k-1\}} \cdot F_k^T + Q_k \end{aligned}$$

where $F_k = df/dx|_{x_{\{k-1\}}}$ is the Jacobian (7×7), Q_k = process noise covariance.

Observation vector (INSAT LST + SMOS soil moisture + ATAL BT):

$$\begin{aligned} y_k &= [T_s_obs, \text{theta_1_obs}, BT_6.2um_obs]^T \\ H_k &= \begin{bmatrix} [1, 0, 0, 0, 0, 0, 0], & \text{[observes } T_s] \\ [0, 1, 0, 0, 0, 0, 0], & \text{[observes } \text{theta_1}] \\ [0, 0, 0, 0, 0, 0, 0] & \text{[UTLS BT handled via ATAL module]} \end{bmatrix} \end{aligned}$$

Update step:

$$\begin{aligned} K_k &= P_{\{k|k-1\}} \cdot H_k^T \cdot (H_k \cdot P_{\{k|k-1\}} \cdot H_k^T + R_k)^{-1} \\ x_hat &= x_hat_{\{k|k-1\}} + K_k \cdot (y_k - H_k \cdot x_hat_{\{k|k-1\}}) \\ P_k &= (I - K_k \cdot H_k) \cdot P_{\{k|k-1\}} \end{aligned}$$

$R_k = \text{diag}([1.5 \text{ K}^2, 0.001 \text{ m}^6/\text{m}^6, 0.5 \text{ K}^2])$ for LST, SM, and BT noise.

Hypersensitive Regime Detection (post-update diagnostic):

```
if theta_1 < theta_pwp AND theta_2 < theta_pwp AND theta_3 < theta_pwp:
    HRI = 1
    raise HeatwaveAlert(severity = (T_s - T_clim) / sigma_T)
```

3D Vertical Soil BEC (separable formulation, per §2.6):

$$B_soil(z_i, z_j) = B_horiz \cdot \exp(-|z_i - z_j| / L_z) \quad [L_z = 0.5\text{m}]$$

This propagates deep-layer drought signals upward to correct surface H and LE estimates.

4.2.2 Hybrid ETKF-3DVAR — Atmospheric State

3DVAR Cost Function:

$$\begin{aligned} J(x) &= 0.5 \cdot (x - x_b)^T \cdot B^{-1} \cdot (x - x_b) \\ &+ 0.5 \cdot (y_o - H(x))^T \cdot R^{-1} \cdot (y_o - H(x)) \end{aligned}$$

- x_b = background (short-range forecast), B = background error covariance
- y_o = observations, $H(x)$ = forward observation operator, R = observation error covariance

Hybrid Background Error Covariance:

$$B_{\text{hybrid}} = \beta_1 \cdot B_{\text{static}} + \beta_2 \cdot P_e$$

- B_{static} = climatological (historical) BEC matrix
- P_e = flow-dependent ensemble covariance from ETKF: $P_e = (1/(N-1)) \cdot X_e \cdot X_e^T$
- X_e = $N \times n$ matrix of ensemble perturbations, $N=50$, $\beta_1 + \beta_2 = 1$, $\beta_1 \sim 0.4$

ETKF Analysis Update:

$$\begin{aligned} X_a &= X_b \cdot T_a && \text{[ensemble transform]} \\ T_a &= [(N-1) \cdot I + Y_b^T \cdot R^{-1} \cdot Y_b]^{-1/2} && \text{[transform matrix]} \end{aligned}$$

where $Y_b = H \cdot X_b$ = ensemble perturbations in observation space.

Minimization of $J(x)$ via L-BFGS-B. Gradient:

$$\text{grad}_x J = B^{-1}(x - x_b) - H^T \cdot R^{-1} \cdot (y_o - H(x))$$

4.3 Layer C — AI Surrogate Models

4.3.1 CNN-FFT Denoising

Signal decomposition using FFT before spatial convolution:

$$\begin{aligned} X_{\text{tilde}}(w) &= \text{FFT}(X) \\ X_{\text{tilde_filtered}}(w) &= X_{\text{tilde}}(w) \cdot G(w) && \text{[Gaussian low-pass: } G(w) = \exp(-w^2 / 2 \cdot \sigma^2)] \\ X_{\text{clean}} &= \text{IFFT}(X_{\text{tilde_filtered}}) + \text{CNN_residual}(X) \end{aligned}$$

The CNN_residual block uses 3 layers of Conv(3×3) -> BatchNorm -> ReLU to capture spatial structure missed by global filtering.

4.3.2 Moran's I — Spatial Autocorrelation Weight

For a spatial field z over grid points i, j :

$$I = (N / S_0) \cdot (\text{Sum}_i \text{Sum}_j w_{ij} \cdot (z_i - z_{\text{bar}})(z_j - z_{\text{bar}})) / (\text{Sum}_i (z_i - z_{\text{bar}})^2)$$

where w_{ij} = spatial weight matrix (inverse distance squared), $S_0 = \text{Sum}_i \text{Sum}_j w_{ij}$.

I in $[-1, 1]$: High I means spatially clustered anomaly -> high attention weight for SARITA.

4.3.3 SARITA — Deformable ConvLSTM Architecture

Standard ConvLSTM equations:

$$i_t = \text{sigmoid}(W_{xi} \cdot X_t + W_{hi} \cdot H_{t-1} + W_{ci} \cdot C_{t-1} + b_i)$$

```
f_t = sigmoid(W_xf * X_t + W_hf * H_{t-1} + W_cf o C_{t-1} + b_f)
C_t = f_t o C_{t-1} + i_t o tanh(W_xc * X_t + W_hc * H_{t-1} + b_c)
o_t = sigmoid(W_xo * X_t + W_ho * H_{t-1} + W_co o C_t + b_o)
H_t = o_t o tanh(C_t)
```

Deformable Convolution extension (replaces standard convolution *):

```
y(p_0) = Sum_{k=1}^{K} w_k · x(p_0 + p_k + Delta_p_k)
```

where p_k = regular grid offsets, Δp_k = learned deformation offsets (predicted by auxiliary conv layer), K = kernel size (e.g. $3 \times 3 = 9$).

Moran-weighted spatial attention gate:

```
A = softmax(I_spatial / tau) [tau = temperature parameter]
H_t_attended = A elementwise-multiply H_t
```

SARITA Input/Output:

- Input: X_t in $R^{T \times H \times W \times C}$ — $T=14$ time steps, $H \times W = 4954$ grid, $C = \{RF, T_{max}, T_{min}, LST, SM, u, v\} = 7$ channels
- Output: $\hat{Y}$ in $R^{7 \times H \times W \times 7}$ — 7-day forecast, 7 variables

4.3.4 Physics-Informed Neural Network (PINN) — What-If Engine [UPGRADED]

Network maps $(x, y, p, t, \text{theta_scenario}) \rightarrow (u, v, T, q, RF, AOD, ATAL_eps)$, where theta_scenario encodes user slider values (UHI, deforestation, aerosol loading).

Fourier Feature Encoding for Monsoon Seasonality [NEW]:

Raw time input t is promoted to a high-dimensional sinusoidal basis that mathematically locks the network to India's annual monsoon cycle:

```
gamma_time(t) = [sin(2pi·t/T_1), cos(2pi·t/T_1),
                  sin(2pi·t/T_2), cos(2pi·t/T_2),
                  sin(2pi·t/T_3), cos(2pi·t/T_3), ...]
```

Frequency set: $T_1 = 365.25$ days (annual), $T_2 = 182.6$ days (semi-annual, onset/withdrawal), $T_3 = 30.4$ days (intraseasonal MJO), $T_4 = 7$ days (synoptic), and random Fourier features B in $R^{256 \times 5}$ for spatial coords.

This prevents the AI from treating the monsoon onset jump (May→June) as "unexplained variance" and forces physically meaningful seasonal representations. Combined with STL decomposition embeddings, the network explicitly separates trend, seasonal, and residual components.

Extended Total Loss Function:

```
L(theta) = L_data + lambda_1·L_PDE + lambda_2·L_BC + lambda_3·L_IC
           + lambda_4·L_RH + lambda_5·L_aerosol + lambda_6·L_ATAL
```

Data loss (against assimilated state):

```
L_data = (1/N_d) Sum_i || y_hat_i - y_i ||^2
```

PDE residual loss (Advection-Diffusion for moisture q):

$$L_{\text{PDE}} = (1/N_c) \sum_j || dq_{\text{hat}}/dt + u_{\text{hat}} \cdot dq_{\text{hat}}/dx + v_{\text{hat}} \cdot dq_{\text{hat}}/dy - K_h \cdot \text{Laplacian}(q_{\text{hat}}) - S_{q_{\text{hat}}} ||^2$$

Continuity residual:

$$L_{\text{div}} = (1/N_c) \sum_j || du_{\text{hat}}/dx + dv_{\text{hat}}/dy + dw_{\text{hat}}/dp ||^2$$

Boundary conditions (topographic mask — zero flux through Himalayan barrier):

$$L_{\text{BC}} = (1/N_b) \sum_k || n_{\text{hat}} \cdot \text{grad}(q_{\text{hat}}) ||^2 \quad [\text{at terrain boundary nodes}]$$

Explicit Relative-Humidity Constraint Loss [NEW] (prevents spurious rainfall in dry air):

$$RH_{\text{hat}} = (e_{\text{hat}} / e_s(T_{\text{hat}})) \cdot 100 \quad [\text{diagnosed from } q_{\text{hat}} \text{ and } T_{\text{hat}} \text{ via Clausius-Clapeyron}]$$

$$L_{\text{RH}} = (1/N_c) \sum_j \max(0, RF_{\text{hat}_j})^2 \cdot \text{Indicator}(RH_{\text{hat}_j} < RH_{\text{threshold}})$$

- $RH_{\text{threshold}} = 60\%$ — physically, rainfall requires near-saturated conditions
- The `Indicator` function is implemented as a soft sigmoid gate during training:

$$w_{\text{RH}}(j) = \text{sigmoid}(-(RH_{\text{hat}_j} - 60) / \tau_{\text{RH}}) \quad [\tau_{\text{RH}} = 5\%]$$
$$L_{\text{RH}} = (1/N_c) \sum_j w_{\text{RH}}(j) \cdot RF_{\text{hat}_j}^2$$

- Effect: any predicted rainfall at low humidity is **penalized quadratically**, driving the network to predict near-zero RF below the threshold.

Aerosol-Lightning-Rainfall Coupling Loss [NEW]:

$$F_{\text{hat_lightning}} = a \cdot W_{\text{hat_max}}^b \cdot \exp(c \cdot AOD_{\text{hat}}) \cdot \text{sigmoid}((RH_{\text{hat}} - 70) / 10)$$

$$L_{\text{aerosol}} = (1/N_c) \sum_j || F_{\text{hat_lightning}_j} - F_{\text{obs_lightning}_j} ||^2 + \mu \cdot || r_{e_hat} - r_{e_target}(AOD_{\text{hat}}) ||^2$$

where $r_{e_target}(AOD) = r_{e0} \cdot AOD^{\{-1/3\}}$ (Twomey's relation), encouraging physically consistent droplet sizing.

ATAL Radiation Forcing Loss [NEW]:

$$L_{\text{ATAL}} = (1/N_b) \sum_k || R_{n_hat_k} - (R_{n_clear} - dF_{\text{SW}}(AOD_{\text{ATAL_hat}}) - dF_{\text{LW}}(\epsilon_{\text{ATAL_hat}})) ||^2$$

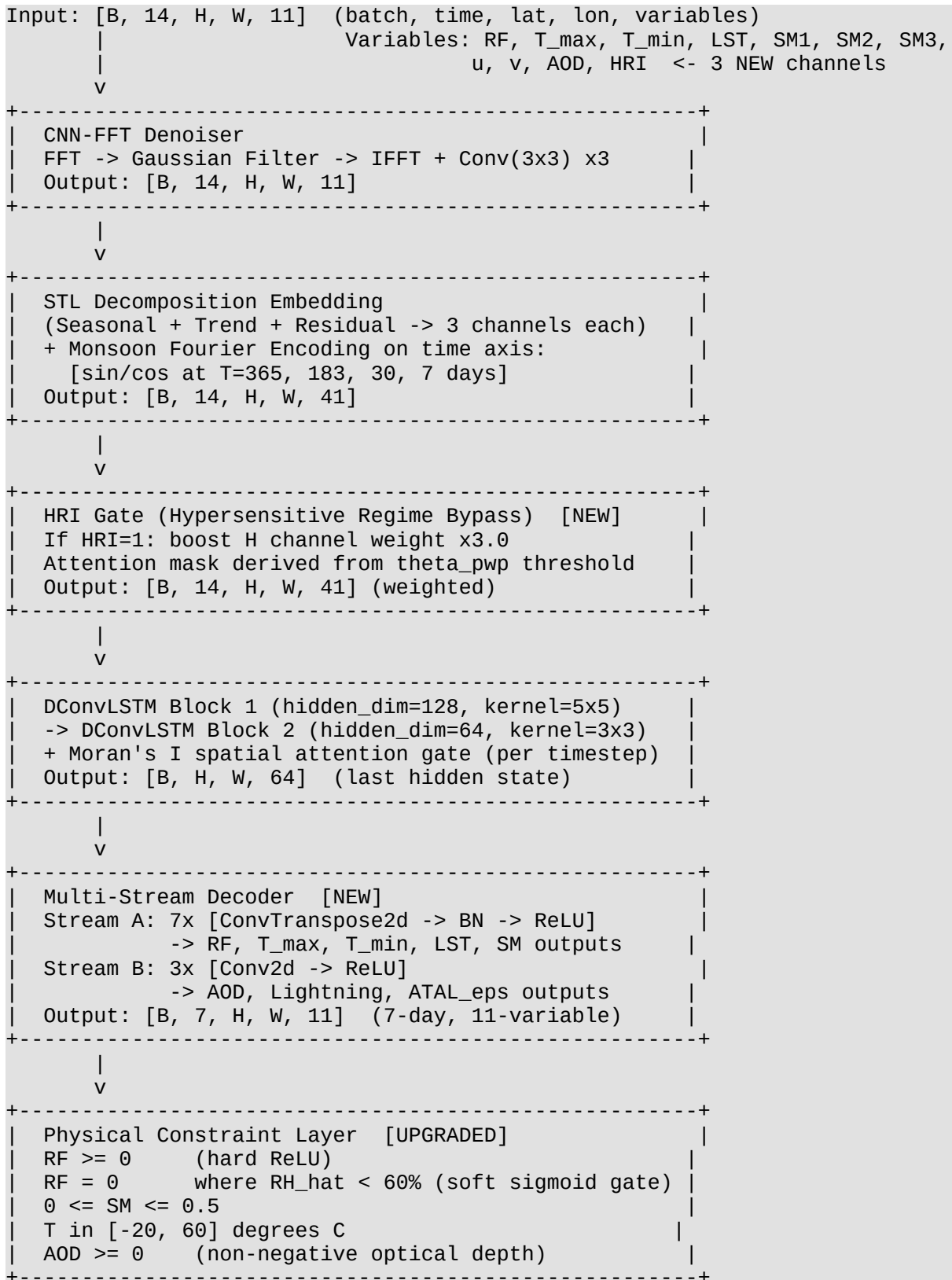
Forces the network surface energy balance to account for the ATAL shortwave and longwave corrections computed from INSAT-3D sounder retrievals.

Derivatives d/dx , d/dt , `Laplacian` are computed via **automatic differentiation** (PyTorch `autograd`).

Lambda tuning schedule: | Loss Term | lambda value | Notes | |---|---|---| | L_{data} | 1.0 | Anchors to observations | | L_{PDE} (Adv-Diff) | 0.1 | Gradually increased during training | | L_{div} (continuity) | 1.0 | Hard physical constraint | | L_{BC} (topography) | 1.0 | Hard boundary | | L_{RH} (humidity gate) | 5.0 | High weight: physically critical | | L_{aerosol} | 0.5 | Tuned against IITM lightning climatology | | L_{ATAL} | 0.3 | Tuned against INSAT-3D sounder BT |

5. AI Model Architecture Details

5.1 SARITA Network Topology [UPGRADED]



Parameters: ~18M trainable weights. Training compute: ~12 GPU-hours on A100.

5.2 PINN Architecture [UPGRADED]

Input: $[x, y, p, t, \text{theta_scenario}, \text{AOD}, \text{HRI_flag}]$ in $\mathbb{R}^8$

```
+-----+
| Monsoon Fourier Feature Encoding [UPGRADED] |
+-----+
| Spatial features (x, y, p):                  |
|   gamma_s = [sin(2pi·B_s·v), cos(2pi·B_s·v)] |
|   B_s in  $\mathbb{R}^{\{128 \times 3\}}$  ~ N(0,  $10^2$ ) |
+-----+
| Temporal features (t):                      |
|   gamma_t = [sin(2pi·t/365.25),              |
|               cos(2pi·t/365.25),  <- annual |
|               sin(2pi·t/182.6),   <- semi-ann |
|               cos(2pi·t/182.6),   <- semi-ann |
|               sin(2pi·t/30.4),    <- MJO      |
|               cos(2pi·t/30.4),    <- MJO      |
|               sin(2pi·t/7),       <- synoptic |
|               cos(2pi·t/7)]       <- synoptic |
+-----+
| Scenario features (theta, AOD, HRI):         |
|   Linear embedding ->  $\mathbb{R}^{64}$            |
+-----+
| Concatenated output:  $\mathbb{R}^{640}$            |
+-----+
|
+-----+
| 8x [Linear(640) -> SiLU]                     |
| + skip connections every 2 layers            |
| (SiLU avoids spectral bias vs ReLU)         |
+-----+
|
+-----+
| Output heads (separate Linear layers):      |
| u, v, T, q ->  $\mathbb{R}$  (dynamical fields)      |
| RF ->  $\mathbb{R}^+$  (via Softplus)              |
| AOD ->  $\mathbb{R}^+$  (aerosol optical depth)      |
| ATAL_eps -> [0,1] (ATAL emissivity, sigmoid)|
| F_lightning ->  $\mathbb{R}^+$  (flash rate, Softplus)|
+-----+
```

Collocation points: 100K interior (LHS) + 10K boundary + 5K UTLS points (for ATAL). **Optimizer:** Adam (lr=1e-3, 1000 epochs) -> L-BFGS fine-tuning (500 iterations).

6. Advanced Research Modules

6.1 Module R1 — Soil Moisture Hypersensitive Regime Monitor

Trigger condition (from §2.6):

```
# Computed every 30 minutes post-SEKF update
def check_hri(theta_1, theta_2, theta_3, theta_pwp=0.14):
    if all([t < theta_pwp for t in [theta_1, theta_2, theta_3]]):
        return True, compute_heatwave_severity()
    return False, 0.0
```

Dashboard output: A grid map showing HRI=1 zones overlaid with T_{max} anomaly. Alert thresholds:

- Yellow: HRI=1, T_{max} anomaly $> 2^{\circ}\text{C}$
- Orange: HRI=1, T_{max} anomaly $> 4^{\circ}\text{C}$, persists > 3 days
- Red: HRI=1, T_{max} anomaly $> 5^{\circ}\text{C}$ — subseasonal heatwave onset imminent

6.2 Module R2 — Relative-Humidity Rainfall Guard

The RH-RF constraint (from §4.3.4 L_RH) is additionally enforced as a **hard post-processing gate** after SARITA inference:

```
RH_850hPa = compute_RH(q_hat, T_hat, p=850)    # 850 hPa level
mask_dry   = RH_850hPa < 60.0                  # Boolean mask
RF_hat[mask_dry] = 0.0                         # Zero rainfall in dry air
```

This dual enforcement (soft loss during training + hard gate at inference) guarantees physically trustworthy forecasts for arid/semi-arid regions of Rajasthan and the Deccan Plateau.

6.3 Module R3 — Aerosol-Lightning-Rainfall Non-Linear Coupler

Data inputs: INSAT-3D L1C visible + IR channels (AOD retrieval), IITM Lightning Imaging Sensor (LIS) archive, CPCB PM2.5 station network.

AOD Retrieval (from INSAT-3D visible reflectance):

```
tau_AOD = -ln(rho_obs / rho_Rayleigh) / (2 / cos(theta_sat))
```

where ρ_{obs} = observed reflectance, ρ_{Rayleigh} = molecular scattering baseline.

Non-linear coupling logic:

- If $\text{AOD} > 0.4$ AND $\text{RH} > 70\%$: aerosol invigoration regime -> CCN-enhanced updrafts -> CG+ lightning burst predicted
- If $\text{AOD} > 0.4$ AND $\text{RH} < 60\%$: aerosol-inhibition regime -> suppressed convection -> rainfall suppression
- This bifurcation is parameterized via the $g(\text{AOD}, \text{RH})$ function in §2.7

Dashboard output: Tab 3 "Aerosol-Storm Link" panel showing AOD heatmap, lightning flash rate forecast, and RH-AOD phase diagram for current grid cell.

6.4 Module R4 — Fourier Monsoon Seasonality Encoder

Beyond PINN input encoding (§4.3.4), the SARITA training pipeline also includes a **learnable monsoon calendar embedding**:

```
class MonsoonSeasonEmbed(nn.Module):
    def __init__(self, d_model=64):
        super().__init__()
        self.freqs = [365.25, 182.6, 30.4, 7.0] # days
        self.proj = nn.Linear(len(self.freqs)*2, d_model)

    def forward(self, t_day_of_year):
        features = []
        for T in self.freqs:
            features += [torch.sin(2*pi*t/T), torch.cos(2*pi*t/T)]
        return self.proj(torch.stack(features, dim=-1))
```

This embedding is concatenated with spatial features before the DConvLSTM layers, ensuring the model's long-range temporal patterns are dominated by physical monsoon frequencies rather than learned spuriously from noise.

6.5 Module R5 — ATAL Dynamics Monitor

INSAT-3D Sounder channels used: | Channel | Wavelength | Sensitivity | ATAL variable derived |
| ---|---|---|---| | WV1 | 6.2 μm | Upper-tropospheric WV | UTLS temperature profile | | WV2 | 7.1 μm | Mid-tropospheric WV | Anticyclone boundary height | | IR1 | 10.8 μm | Cloud-top temperature | Convective outflow altitude |

ATAL emissivity retrieval:

```
BT_6.2_clear = RTM(T_profile_ERA5, q_profile_ERA5)    [clear-sky reference BT]
delta_BT = BT_6.2_observed - BT_6.2_clear
epsilon_ATAL = 1 - exp(-tau_ATAL)    where tau_ATAL = f(delta_BT,
viewing_angle)
```

Dashboard output: Vertical cross-section panel (pressure vs longitude at 25°N) showing ATAL emissivity, anticyclone boundaries (350K isentrope), and trapped tracer concentration anomalies.

7. Data Schema & Infrastructure

7.1 Input Data Sources [UPGRADED]

Source	Variable	Format	Cadence	Resolution
IMD Gridded	RF, T_max, T_min	Binary .grd	Daily	27km
INSAT-3DR L2B	LST, Precipitation, Cloud Mask	HDF5	30-min	4km
INSAT-3D L1C	Multispectral radiance	HDF5	30-min	1km
INSAT-3D Sounder	WV BT (6.2μm, 7.1μm), IR (10.8μm)	HDF5	30-min	10km
ERA5 (ECMWF)	u, v, w, RH, Geopotential	NetCDF4	Hourly	31km
IMD AWS	Station T, RH, Wind	CSV	Hourly	Point
MOSDAC Soil	Soil Moisture 3-layer (SMOS/SCATSA T)	HDF5	Daily	25km
IITM Lightning	CG+ flash rates, stroke density	NetCDF4	Hourly	0.5°
CPCB AQI	PM2.5, PM10,	CSV	Hourly	Station

Source	Variable	Format	Cadence	Resolution
	AOD proxy			

7.2 Processed DataCube Schema [UPGRADED]

```
xarray.Dataset {
  dimensions: (time: unbounded, lat: 3200, lon: 2900, level: 17, soil_layer: 3)

  variables:
    RF          (time, lat, lon)          float32  [mm/day]
    T_max       (time, lat, lon)          float32  [degrees C]
    T_min       (time, lat, lon)          float32  [degrees C]
    LST         (time, lat, lon)          float32  [K]
    SM          (time, soil_layer, lat, lon) float32  [m3/m3]  <- 3 layers
  [NEW]
    HRI         (time, lat, lon)          uint8    [0 or 1]  <- [NEW]
    u_wind      (time, level, lat, lon)    float32  [m/s]
    v_wind      (time, level, lat, lon)    float32  [m/s]
    RH          (time, level, lat, lon)    float32  [%]
    AOD         (time, lat, lon)          float32  [dimensionless]  <-
  [NEW]
    F_lightning (time, lat, lon)          float32  [flashes/km2/hr]  <-
  [NEW]
    ATAL_eps    (time, lat, lon)          float32  [0-1]    <- [NEW]
    BT_6p2um    (time, lat, lon)          float32  [K]      <- [NEW]

  attributes:
    CRS: EPSG:4326
    conventions: CF-1.8
    spatial_resolution: 4km
    fill_value: NaN
    soil_depths_m: [0.0-0.1, 0.1-0.4, 0.4-1.0]
}
```

7.3 Storage Architecture

```
data_lake/
+-- raw/
|   +-- imd/          -> .grd binary files (partitioned by year/month)
|   +-- insat/        -> HDF5 files (partitioned by year/month/day)
|   |   +-- l2b/       -> LST, precipitation, cloud mask
|   |   +-- sounder/   -> WV BT channels (6.2, 7.1 μm)
|   +-- era5/         -> NetCDF4 (partitioned by year/month)
|   +-- lightning/    -> IITM LIS flash archive (NetCDF4)
|   +-- aerosol/      -> CPCB AQI CSVs + derived AOD
|
+-- harmonized/
|   +-- datacube/     -> Zarr chunked arrays (chunk: 24h x 200lat x 200lon)
|
+-- assimilated/
|   +-- analysis/     -> Optimal state x_a per 6-hour cycle (NetCDF4)
|
+-- forecasts/
|   +-- ai_output/    -> SARITA 7-day forecasts (COGeoTIFF per variable)
|
+-- scenarios/
|   +-- pinn_runs/    -> PINN what-if outputs keyed by scenario hash
```

Query Engine: DuckDB with spatial extension for sub-second SQL over Parquet/Zarr slices. **File**

format for serving: Cloud-Optimized GeoTIFF (COG) — enables HTTP range requests for partial reads.

8. Technology Stack

Category	Library	Version	Purpose
Data I/O	imdlb	>=0.1.8	IMD .grd binary ingestion
Data I/O	h5py	>=3.9	INSAT HDF5 + Sounder parsing
Data I/O	cdsapi	>=0.6	ERA5 download via CDS API
Geospatial	rioxarray	>=0.15	Reprojection, CRS transform
Geospatial	pyproj	>=3.6	Coordinate transforms
Geospatial	geopandas	>=0.14	Vector overlays
Arrays	xarray	>=2023.9	N-D labeled arrays, CF
Arrays	zarr	>=2.16	Chunked array storage
Statistics	pysal	>=2.5	Moran's I computation
Statistics	statsmodels	>=0.14	STL decomposition
ML	torch	>=2.1	SARITA, PINN training
ML	scipy	>=1.11	FFT, L-BFGS-B
ML	jax	>=0.4	sEKF Jacobian autodiff
DA	filterpy	>=1.4	sEKF implementation
Query	duckdb	>=0.9	Vectorized SQL analytics
Viz	streamlit	>=1.28	Dashboard frontend
Viz	plotly	>=5.17	Interactive maps/plots
Viz	folium	>=0.15	Leaflet.js tiled maps
Serving	rasterio	>=1.3	COGeoTIFF read/write
Serving	fastapi	>=0.104	REST API for PINN inference
Aerosol	py-RRTMG	>=1.0	Radiative transfer for ATAL forcing
Aerosol	pytmatrix	>=0.3	Mie scattering for AOD retrieval

9. Phased Implementation Plan

Phase 1 — Foundation & Data Engineering (Months 1–3)

Goal: Reproducible, harmonized DataCube ready for training.

Month 1 — IMD + Lightning + Aerosol Pipelines

- ☐ Install `imdlib`, write `ingest_imd.py`: fetch RF, T_max, T_min .grd for 1901–present
- ☐ Replace fill values (-999.0) -> NaN; convert to CF-compliant `xarray.Dataset`
- ☐ Chunk and write to `data_lake/raw/imd/` as Zarr
- ☐ Implement Kriging interpolator for missing station data
- ☐ **[R3]** Write `ingest_lightning.py`: parse IITM LIS flash-rate archive (NetCDF4) -> 0.5° grid
- ☐ **[R3]** Write `ingest_aerosol.py`: parse CPCB PM2.5 CSV + compute AOD proxy via Angstrom exponent
- ☐ Unit tests: mass conservation check (RF budget non-negative)

Month 2 — INSAT Pipelines (L2B + Sounder)

- ☐ Write `ingest_insat.py`: parse 3RIMG_L2B_LST HDF5 via `h5py`
- ☐ Extract LST, Precipitation Estimate, Cloud Mask datasets
- ☐ **[R5]** Write `ingest_sounder.py`: parse INSAT-3D Sounder HDF5 for WV BT channels (6.2μm, 7.1μm, 10.8μm)
- ☐ **[R5]** Implement `atal_retrieval.py`: compute `delta_BT` vs RTM clear-sky baseline -> `epsilon_ATAL`
- ☐ Reproject GEOS -> EPSG:4326 via `rioxarray.reproject()`; regrid 4km/10km -> 27km
- ☐ Set up MOSDAC automated download cron (30-min cadence)
- ☐ Validation: RMSE < 0.5°C between collocated T_max vs LST daytime values

Month 3 — Harmonized DataCube with 11-variable Schema

- ☐ Merge all sources into unified `xarray.Dataset` (CF-1.8), 3 soil layers, AOD, lightning, `ATAL_eps`
- ☐ ERA5 download pipeline (`cdsapi`) — 500hPa u, v, geopotential, RH profile
- ☐ STL decomposition on full IMD time series
- ☐ **[R4]** Compute Monsoon Fourier features for full historical record and store as auxiliary Zarr array
- ☐ Write `build_datacube.py` — final Zarr output with DuckDB index (11 variables)

- **Deliverable:** `datacube_v1.zarr` covering 1990–present, 4km, 11 variables
-

Phase 2 — Data Assimilation Pipeline (Months 4–6)

Goal: Real-time optimal state estimation running on a 6-hour analysis cycle.

Month 4 — *sEKF with 3D Soil + ATAL Observations*

- ☐ **[R1]** Define extended NOAH-LSM forward model $f(x)$ with 3 soil layers (0-10cm, 10-40cm, 40-100cm)
- ☐ **[R1]** Code `sekf.py` using `filterpy.kalman.ExtendedKalmanFilter` with 7-element state vector
- ☐ **[R5]** Extend observation vector to include INSAT-3D Sounder BT_6.2um for UTLS correction
- ☐ **[R1]** Implement `B_soil` with vertical separable covariance (exponential decay, $L_z=0.5m$)
- ☐ Jacobian F_k via JAX autodiff (`jax.jacfwd`)
- ☐ Tune Q_k and $R_k = \text{diag}([1.5, 0.001, 0.5])$ via OSE
- ☐ **[R1]** Implement `check_hri()` post-update diagnostic and HeatwaveAlert severity scoring
- ☐ Validation: 3-layer SM analysis vs SMOS (target: $R^2 > 0.7$ each layer)

Month 5 — *3DVAR Baseline*

- ☐ Implement static BEC matrix `B_static` from 30-year NWP climatological statistics
- ☐ Forward observation operator $H(x)$ for: surface stations, radiosonde profiles, satellite BT, ATAL epsilon
- ☐ 3DVAR cost function minimization via `scipy.optimize.minimize(method='L-BFGS-B')`
- ☐ OSE: verify that analysis x_a has lower 24h forecast error than background x_b

Month 6 — *ETKF + Hybrid*

- ☐ Generate $N=50$ ensemble perturbations using bred vectors
 - ☐ Implement ETKF transform matrix T_a computation
 - ☐ Tune $\beta_1=0.4$, $\beta_2=0.6$ via cross-validation on withheld 2020–2021 monsoon data
 - **Deliverable:** `assimilation_cycle.py` — 6-hourly optimal state output (11 variables) to `data_lake/assimilated/`
-

Phase 3 — AI Model Training & Calibration (Months 7–10)

Goal: Trained SARITA model with validated skill scores.

Month 7 — Data Preparation for Training (11-variable)

- ☐ Build ML-ready dataset: sliding windows of 14 days input -> 7 days target (11 variables)
- ☐ Train/val/test split: 1990–2018 / 2019–2021 / 2022–present
- ☐ Normalize per-variable using climatological mean/std (z-score)
- ☐ Implement Moran's I computation per grid timestep (batch via `pysal.lib.weights`)
- ☐ **[R1]** Label historical HRI=1 episodes using 3-layer SMOS SM vs `theta_pwp` threshold
- ☐ **[R3]** Co-register IITM lightning flash rate with AOD and RH at same grid/time
- ☐ **[R4]** Compute and store Monsoon Fourier feature embeddings for all training timestamps

Month 8 — CNN-FFT Denoiser + HRI Gate

- ☐ Implement `CNNFFTDenoiser` in PyTorch: FFT -> Gaussian spectral filter -> IFFT + residual CNN
- ☐ Train standalone on noisy -> clean pairs (synthetic noise injection)
- ☐ **[R1]** Implement `HRIAttentionGate` module: boosts H-channel weights when HRI=1
- ☐ Metric target: SNR improvement > 5 dB

Month 9 — SARITA Training (11-variable + Monsoon Encoder)

- ☐ Implement `DeformableConv2d` layer (DCNv2-style, learnable offsets)
- ☐ Implement `DConvLSTM` cell with Moran attention gate
- ☐ **[R4]** Implement `MonsoonSeasonEmbed` module and integrate into SARITA input pipeline
- ☐ **[R2]** Add RH-RF hard gate to SARITA Physical Constraint Layer (post-decoder)
- ☐ Full SARITA training: 2x `DConvLSTM` blocks + multi-stream decoder (11 outputs) + physical clipping
- ☐ Loss: $\text{MAE} + 0.1 \cdot \text{SSIM_loss} + \text{aerosol coupling auxiliary loss on AOD and F_lightning heads}$
- ☐ Training: 100 epochs, batch=8, AdamW (lr=3e-4, weight_decay=1e-5), cosine LR decay

Month 10 — Benchmarking

- ☐ Evaluate on 2022–present test set: target $\text{RMSE(RF)} < 5 \text{ mm/day}$ vs IMD observed
- ☐ Target CSI (Critical Success Index) for extreme events ($>100\text{mm/day}$) > 0.40
- ☐ **[R1]** Evaluate HRI: compare predicted heatwave onset day vs observed (target: $\text{MAE} < 1.5 \text{ days}$)

- ☐ [R3] Evaluate lightning forecast: CSI vs IITM archive (target: CSI > 0.35)
 - ☐ [R5] Evaluate ATAL epsilon: compare vs CALIPSO aerosol lidar profiles (target: $R^2 > 0.6$)
 - ☐ Compare vs: persistence, climatology, IMD STEPS-NWP baseline
 - **Deliverable:** Trained `sarita_v1.pt` checkpoint + evaluation report
-

Phase 4 — PINN What-If Module (Months 11–13)

Goal: Interactive scenario engine responding in < 500ms.

Month 11 — Collocation, Fourier Encoding & Extended PINN Inputs

- ☐ Generate 100K interior collocation points (Latin Hypercube Sampling over full 3D domain)
- ☐ 10K boundary points (Himalayan topographic mask, coastline)
- ☐ 5K UTLS collocation points (100–200 hPa) specifically for ATAL forcing residuals
- ☐ [R4] Implement full `MonsoonFourierEncoding` with 4 frequency tiers (annual, semi-annual, MJO, synoptic)
- ☐ [R3] Add AOD and `HRI_flag` to PINN input vector (8-dimensional input)
- ☐ 8-layer SiLU MLP with skip connections + separate output heads for AOD, ATAL_eps, `F_lightning`

Month 12 — Extended PDE Loss Integration

- ☐ Implement automatic differentiation for all PDE residuals via `torch.autograd.grad`
- ☐ [R2] Code `L_RH` loss: soft sigmoid penalty $w_{RH}(j) \cdot RF_hat_j^2$ for $RH < 60\%$
- ☐ [R3] Code `L_aerosol` loss: Twomey droplet radius consistency + lightning flash rate MSE
- ☐ [R5] Code `L_ATAL` loss: surface energy balance residual with ATAL SW/LW forcing
- ☐ Code `pinn_loss.py` with all 7 loss terms; implement lambda schedule (see §4.3.4 table)
- ☐ Two-stage training: Adam (1000 epochs) -> L-BFGS (500 iterations)
- ☐ Verify: `L_RH` eliminates all instances of $RF > 0$ at $RH < 50\%$ in validation set

Month 13 — Scenario Interface & Neuro-Symbolic Layer

- ☐ Map dashboard sliders -> tensor perturbations `theta_scenario`:
 - UHI intensity slider -> modify `T_s` forcing boundary condition
 - Rainfall anomaly slider -> modify `S_q` source term

- Deforestation slider -> modify surface albedo and roughness z_0
 - **[R3]** Aerosol loading slider -> modify AOD forcing -> predict lightning/RF impact
 - **[R5]** ATAL intensity slider -> modify ϵ_{ATAL} -> compute TOA forcing change
 - ☐ FastAPI endpoint: `POST /scenario` -> runs PINN forward pass -> returns JSON field
 - ☐ Latency target: < 500ms for single scenario on GPU
 - **Deliverable:** `pinn_v1.pt` + REST API server (8-input PINN, 8-output heads)
-

Phase 5 — Dashboard & Deployment (Months 14–15)

Goal: Stakeholder-ready, interactive visualization platform.

Month 14 — Analytical Backend

- ☐ DuckDB schema: index all COGeoTIFF outputs by (`variable`, `date`, `scenario_id`)
- ☐ Write DuckDB SQL views for: time-series at point, spatial average over admin polygon, extreme event detection
- ☐ **[R1]** Write DuckDB view: `heatwave_events` — HRI=1 episodes with duration, severity, spatial extent
- ☐ **[R3]** Write DuckDB view: `aerosol_storm_link` — AOD, RH, $F_{\text{lightning}}$ at co-located grid cells
- ☐ COGeoTIFF generation pipeline (GDAL/rasterio, with overview levels for tiled web maps)

Month 15 — Streamlit Dashboard (8-Tab)

- ☐ **Tab 1 — Live Monitor:** Real-time 30-min INSAT LST heatmap + sEKF-corrected surface fields + HRI alert overlay
- ☐ **Tab 2 — 3D Soil Moisture:** [R1] Three-layer SM viewer (vertical bar chart per grid cell) + θ_{pwp} threshold lines + HeatwaveAlert panel
- ☐ **Tab 3 — AI Forecast:** SARITA 7-day forecast overlay (toggle 11 variables, animation playback)
- ☐ **Tab 4 — What-If Scenarios:** [R2, R3, R5] UHI/deforestation/rainfall/aerosol/ATAL sliders -> PINN diff-maps + probability bars
- ☐ **Tab 5 — Aerosol-Storm Link:** [R3] AOD heatmap + lightning flash rate forecast + RH-AOD phase diagram
- ☐ **Tab 6 — ATAL Monitor:** [R5] UTLS cross-section at 25°N, ϵ_{ATAL} isoline, TOA forcing estimate
- ☐ **Tab 7 — Historical Analytics:** DuckDB-backed time-series explorer, trend lines, STL

display, monsoon onset/withdrawal dates

- ☐ **Tab 8 — Validation:** SARITA skill scores, sEKF analysis increments, OSE results, RH-RF constraint compliance rate
- ☐ UAT + latency testing: dashboard load < 3s, scenario response < 500ms

10. Verification & Benchmarking

10.1 Quantitative Skill Metrics [UPGRADED]

Component	Metric	Target	Baseline
sEKF (LST)	RMSE vs INSAT T _s	< 1.0 K	Open-loop: ~3.5K
sEKF (SM layer 1)	R ² vs SMOS	> 0.70	Open-loop: ~0.45
sEKF (SM layer 2)	R ² vs SMOS	> 0.65	Open-loop: ~0.38
sEKF (SM layer 3)	R ² vs SMOS	> 0.55	Open-loop: ~0.30
[R1] HRI detection	Heatwave onset MAE	< 1.5 days	Climatology: ~4 days
3DVAR analysis	500hPa wind RMS error	< 3 m/s	Background: ~6 m/s
SARITA (RF)	RMSE vs IMD	< 5 mm/day	Persistence: ~12 mm/day
SARITA (extreme)	CSI @ 100mm/day	> 0.40	NWP baseline: ~0.32
[R2] RH-RF guard	RF>0 at RH<50% rate	0%	Unconstrained: ~8%
[R3] Lightning CSI	CSI vs IITM archive	> 0.35	Climatology: ~0.18
[R5] ATAL epsilon	R ² vs CALIPSO lidar	> 0.60	No baseline exists
PINN (PDE residual)	Advection-Diffusion norm	< 1e-3	N/A
PINN (RH constraint)	L _{RH} at convergence	< 1e-4	Unconstrained: ~0.1
PINN (latency)	Inference time	< 500ms	NWP: ~hours

10.2 Observing System Experiments (OSE)

Run parallel analyses denying each observation type in turn:

- Deny INSAT LST -> measure degradation in sEKF T_s and SM layer 1 analysis
- Deny SMOS SM -> measure degradation in HRI detection skill
- Deny ERA5 winds -> measure degradation in 500hPa flow analysis
- Deny IMD station data -> measure degradation in precipitation analysis
- **[R5]** Deny INSAT-3D Sounder BT -> measure degradation in ATAL epsilon and TOA forcing
- **[R3]** Deny IITM lightning -> measure degradation in aerosol-storm coupling skill

10.3 Physical Consistency Checks [UPGRADED]

- **Mass conservation:** Total RF budget must match net water vapor flux divergence
 - **Energy closure:** $R_n - H - LE - G \sim 0$ (residual $< 5 \text{ W/m}^2$)
 - **Non-negative outputs:** $RF \geq 0$ everywhere; zero at $RH < 60\%$ (hard gate)
 - **Clausius-Clapeyron test:** Simulated extreme RF scenarios must satisfy $\Delta RF / \Delta T \sim 7\% / ^\circ\text{C}$
 - **[R1] SM energy coupling:** When $HRI=1$, verify $H / R_n > 0.85$ (dominant sensible heat)
 - **[R3] Twomey consistency:** Predicted r_e must satisfy $r_e \propto AOD^{-1/3}$ ($R^2 > 0.7$)
 - **[R5] ATAL radiative closure:** $dF_{SW} + dF_{LW}$ from ATAL module must reproduce CERES TOA anomalies within 3 W/m^2
-

Appendix A — Repository Structure [UPGRADED]

```
isro_climate_twin/
+-- ingestion/
|   +-- ingest_imd.py
|   +-- ingest_insat.py
|   +-- ingest_sounder.py      # [R5] INSAT-3D Sounder WV/IR BT channels
|   +-- ingest_era5.py
|   +-- ingest_lightning.py    # [R3] IITM LIS flash archive
|   +-- ingest_aerosol.py      # [R3] CPCB PM2.5 + AOD retrieval
|   +-- atal_retrieval.py      # [R5] ATAL epsilon from delta-BT method
|   +-- build_datacube.py      #      11-variable CF DataCube
+-- assimilation/
|   +-- sekf.py                # [R1] 7-state SEKF with 3D soil + HRI
|   +-- var3d.py
|   +-- etkf.py
|   +-- hybrid_da.py
|   +-- heatwave_alert.py      # [R1] HRI threshold monitor & severity scorer
+-- models/
|   +-- denoiser.py            #      CNN-FFT
|   +-- moran_attention.py     #      Moran's I spatial gate
|   +-- dconv_lstm.py          #      Deformable ConvLSTM cell
|   +-- hri_gate.py            # [R1] Hypersensitive Regime attention bypass
|   +-- monsoon_fourier.py     # [R4] MonsoonSeasonEmbed (4 frequencies)
|   +-- sarita.py              #      Full SARITA network (11 variables)
|   +-- pinn.py                #      PINN forward model (8 inputs, 8 outputs)
|   +-- pinn_loss.py           # [R2,R3,R5] Extended 7-term loss function
+-- training/
|   +-- train_sarita.py
|   +-- train_pinn.py
+-- modules/                  # [NEW] Pluggable research modules
|   +-- r1_soil_hri.py          # [R1] 3D SM breakpoint monitor
|   +-- r2_rh_constraint.py     # [R2] RH-RF guard (loss + hard gate)
|   +-- r3_aerosol_lightning.py # [R3] AOD-CCN-lightning coupler
|   +-- r4_monsoon_encoder.py  # [R4] Fourier seasonality embedding
|   +-- r5_atal_dynamics.py    # [R5] ATAL emissivity + TOA forcing
+-- serving/
|   +-- api.py                  #      FastAPI PINN inference endpoint
|   +-- cog_builder.py          #      COGeoTIFF pipeline
+-- dashboard/
```

```

| +-- app.py # Streamlit 8-tab dashboard
| +-- tabs/
| | +-- tab_live.py
| | +-- tab_soil_3d.py # [R1]
| | +-- tab_forecast.py
| | +-- tab_scenarios.py # [R2, R3, R5]
| | +-- tab_aerosol.py # [R3]
| | +-- tab_atal.py # [R5]
| | +-- tab_history.py
| | +-- tab_validation.py
+-- tests/
| +-- test_ingestion.py
| +-- test_assimilation.py
| +-- test_models.py
| +-- test_pinn_constraints.py # [R2] RH-RF constraint compliance tests
+-- configs/
| +-- config.yaml # Hyperparameters, paths, DA weights,
| lambda schedule
+-- requirements.txt

```

Appendix B — Key References

Core Data Assimilation:

1. Evensen, G. (2003). The Ensemble Kalman Filter. *Ocean Dynamics*, 53(4), 343-367.
2. Wang, X. et al. (2008). A Hybrid ETKF-3DVAR Data Assimilation Scheme. *Mon. Weather Rev.*, 136(3), 1179-1196.

AI Architectures: 3. Shi, X. et al. (2015). Convolutional LSTM Network for Precipitation Nowcasting. *NeurIPS 2015*. 4. Dai, J. et al. (2017). Deformable Convolutional Networks. *ICCV 2017*. 5. Raissi, M. et al. (2019). Physics-Informed Neural Networks. *J. Comput. Phys.*, 378, 686-707. 6. Tancik, M. et al. (2020). Fourier Features Let Networks Learn High Frequency Functions. *NeurIPS 2020*. 7. Anselin, L. (1995). Local Indicators of Spatial Association—LISA. *Geographical Analysis*, 27(2), 93-115.

[R1] Soil Moisture Hypersensitive Regime: 8. Seneviratne, S.I. et al. (2010). Investigating soil moisture–climate interactions in a changing climate. *Earth-Science Reviews*, 99(3-4), 125-161. 9. Lorenz, R. et al. (2015). Persistence of heat waves and its link to soil moisture memory. *Geophys. Res. Lett.*, 42(18), 6692-6699. [Journal of Climate] 10. Koster, R.D. et al. (2004). Regions of Strong Coupling Between Soil Moisture and Precipitation. *Science*, 305, 1138-1140.

[R2] RH Constraints in PINN: 11. Srivastava, A. et al. (2023). Physics-informed deep learning framework for rainfall forecasting in arid and diverse climatic regions. *Environ. Modelling & Software*, 168, 105800.

[R3] Aerosol-Lightning-Rainfall Coupling: 12. Rosenfeld, D. et al. (2014). Global observations of aerosol-cloud-precipitation-climate interactions. *Reviews of Geophysics*, 52(4), 750-808. 13. Lal, D.M. et al. (2021). Aerosol-lightning linkages over the Indian monsoon region. *Atmospheric Research*, 252, 105446. [IITM] 14. Twomey, S. (1977). The influence of pollution on the shortwave albedo of clouds. *J. Atmos. Sci.*, 34(7), 1149-1152.

[R4] Fourier Feature Mapping for Seasonality: 15. Yadav, I. et al. (2022). Uncovering hidden temporal profiles in India's environmental cycles using Physics-Informed Neural Networks. *Environmental Pollution*, 305, 119280.

[R5] ATAL Dynamics: 16. Vernier, J.P. et al. (2011). Major influence of tropical volcanic eruptions on the stratospheric aerosol layer. *Geophys. Res. Lett.*, 38(12). [Atmos. Meas. Tech.] 17. Brunamonti, S. et al. (2018). Balloon-borne measurements of temperature, water vapor, ozone and aerosol backscatter on the southern slopes of the Himalayas. *Atmos. Chem. Phys.*, 18, 15937-15957. 18. Pan, L.L. et al. (2016). Bimodal distribution of free tropospheric ozone over the tropical western Pacific revealed by airborne observations. *Geophys. Res. Lett.*, 42, 7820-7828.

Data Sources: 19. ISRO MOSDAC: <https://www.mosdac.gov.in/> 20. IMD Gridded Data: <https://imdpune.gov.in/> 21. IITM Lightning Data: <https://www.tropmet.res.in/> 22. CPCB AQI Portal: <https://cpcb.nic.in/>